1. Use Case Overview

This use case demonstrates a **Retail Inventory Agent** built using **LangChain**, **AlloyDB**, and **Google Cloud services**. The solution automates inventory replenishment by leveraging real-time signals, structured data, and contextual documents. The agentic workflow utilizes **Retrieval-Augmented Generation (RAG)** and decision logic to generate purchase orders and manage stock levels efficiently.

2. Architecture & Components

Key technologies and components used:

- **AlloyDB** with pgvector and **ScaNN** for semantic vector search.
- **LangChain** integrated with AlloyDBVectorStore and SQLDatabaseToolkit.
- **Vertex AI Embeddings** for generating semantic vectors.
- **Google Cloud Services** including GCS, Cloud Run, and Secret Manager.

3. Agent Workflow

The agent follows a **five-step decision-making process**:

1. **Signals & Triggers**
   Detect low stock, demand spikes, or promotional events.
2. **Structured Facts**
   Query AlloyDB for:
   a. Stock levels
   b. Purchase orders
   c. Sales velocity
   d. Lead times
3. **Context via RAG**
   Retrieve:
   a. Supplier policies
   b. Procurement guidelines
   c. Operational SOPs
4. **Plan & Decide**
   a. Compute recommended quantity:
   b. Apply policy constraints (PO caps, MOQ, substitution rules).

recommended_qty = forecast(horizon_days) + safety_stock - net_position

5. **Act & Log**
    a. Generate purchase orders.
    b. Notify approvers.
    c. Log transactions for audit.

4. Data Model

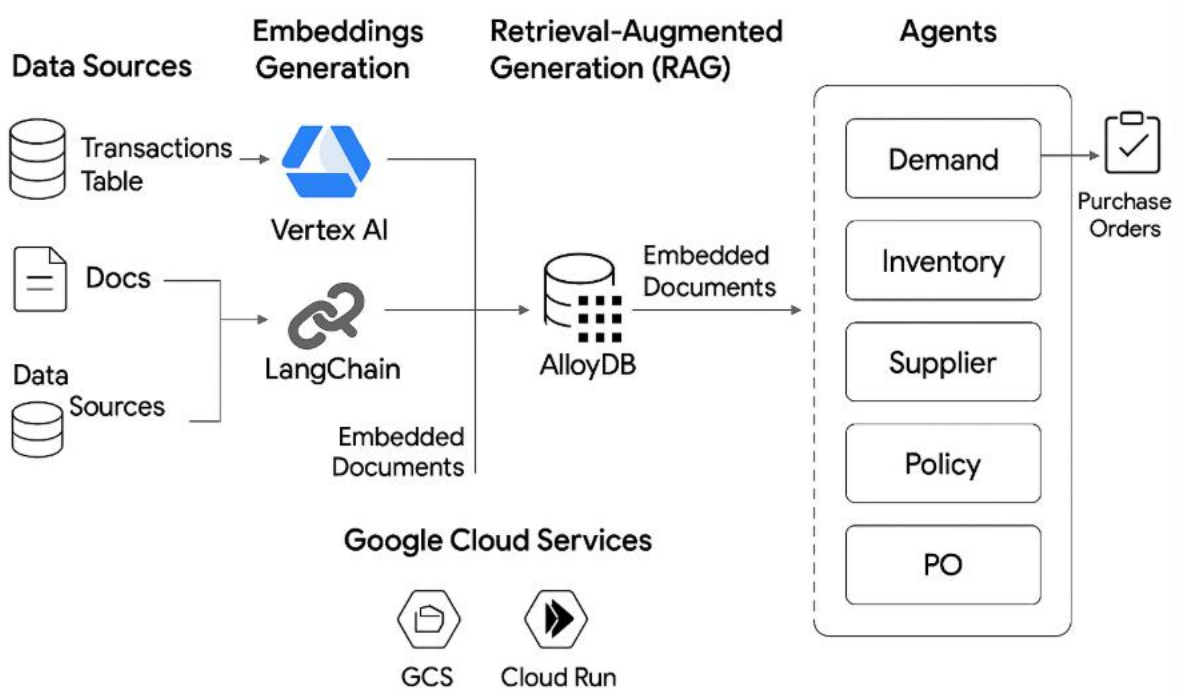
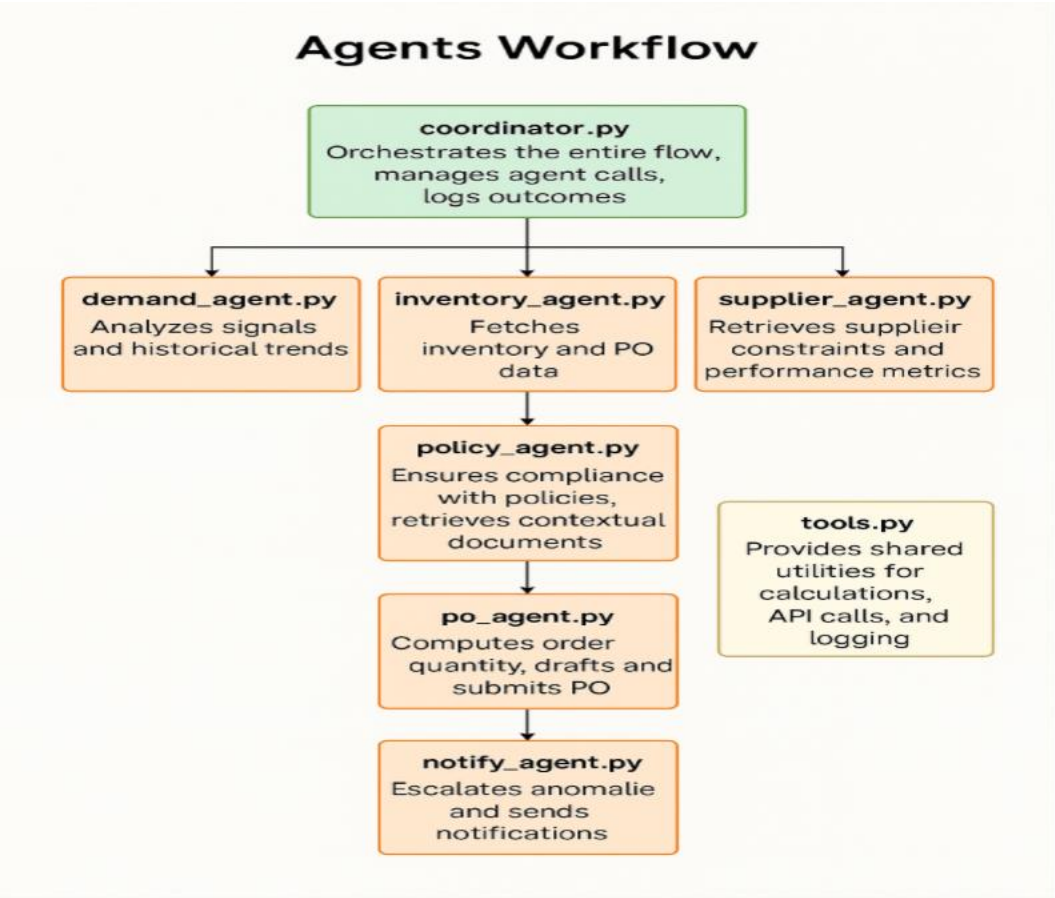
The application uses a minimal set of relational tables:

- transactions(order_id, sku, quantity, price, order_ts, channel, store_id)
- stock_levels(sku, store_id, on_hand, in_transit, updated_at)
- products(sku, title, description, category, embedding VECTOR(768))
- suppliers(supplier_id, lead_time_days, moq, cost, terms)

5. Performance Considerations

- **pgvector + ScaNN** indexing is used for efficient similarity search.
- Embeddings are cached or stored in Alloy DB for faster retrieval.
- **Error Handling & Fallbacks**:
    - The agent falls back to a **7-day moving average** if forecasting fails.
    - Policy constraints (e.g., PO caps, substitution rules) are enforced during decision-making.

6. Architecture Diagram



The following diagram illustrates the end-to-end flow of the Retail Inventory Agent: